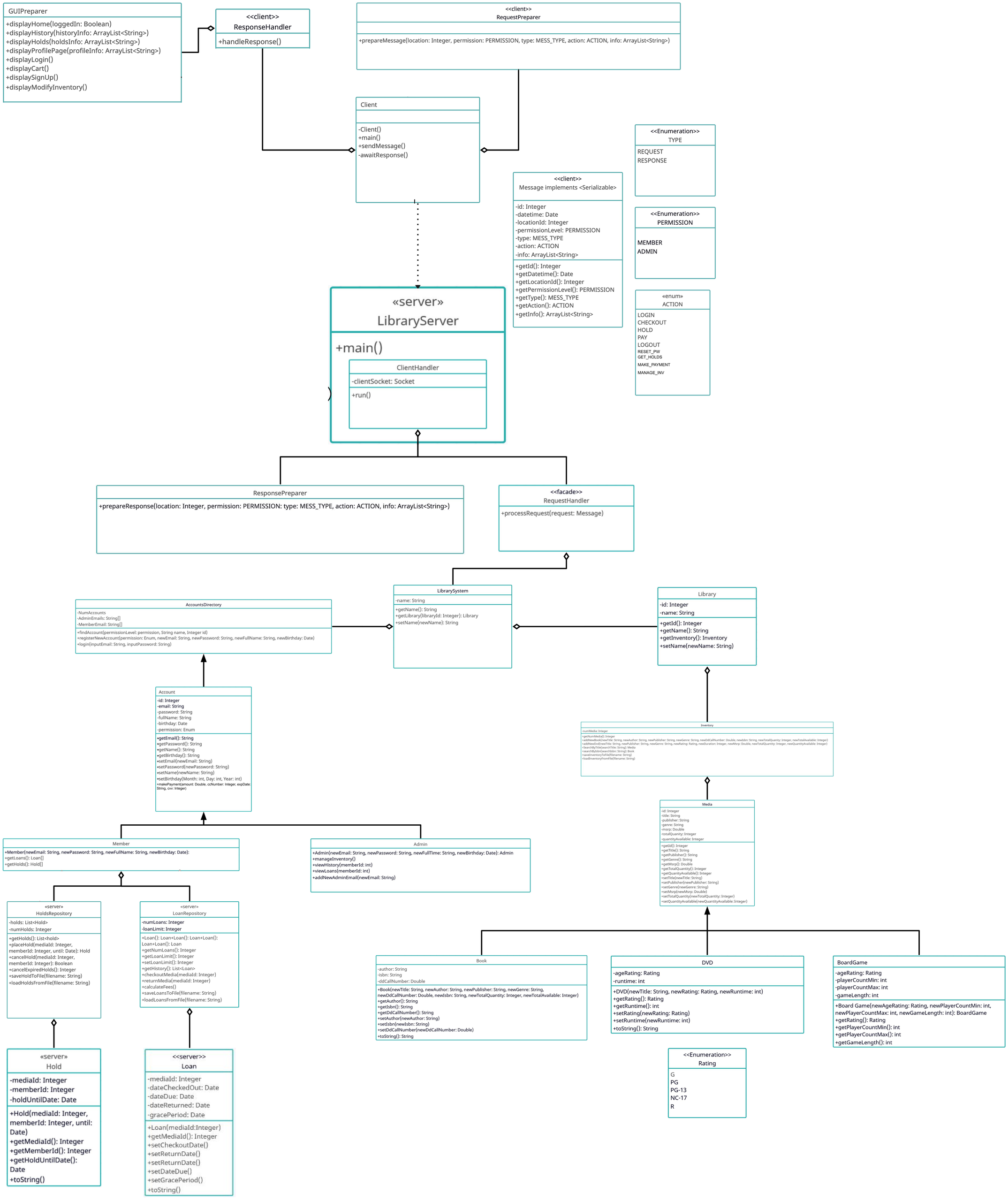
GUIPreparer
+displayHome(loggedIn: Boolean)
+displayHistory(historyInfo: ArrayList<String>)
+displayHolds(holdsInfo: ArrayList<String>)
+displayProfilePage(profileInfo: ArrayList<String>)
+displayLogin()
+displayCart()
+displaySignUp()
+displayModifyInventory()
<<client>>
ResponseHandler
+handleResponse()
<<client>>
RequestPreparer
+prepareMessage(location: Integer, permission: PERMISSION, type: MESS_TYPE, action: ACTION, info: ArrayList<String>)
Client
-Client()
+main()
+sendMessage()
-awaitResponse()
<<Enumeration>>
TYPE
REQUEST
RESPONSE
<<client>>
Message implements <Serializable>
-id: Integer
-datetime: Date
-locationId: Integer
-permissionLevel: PERMISSION
-type: MESS_TYPE
-action: ACTION
-info: ArrayList<String>
+getId(): Integer
+getDatetime(): Date
+getLocationId(): Integer
+getPermissionLevel(): PERMISSION
+getType(): MESS_TYPE
+getAction(): ACTION
+getInfo(): ArrayList<String>
<<Enumeration>>
PERMISSION
MEMBER
ADMIN
«enum»
ACTION
LOGIN
CHECKOUT
HOLD
PAY
LOGOUT
RESET_PW
GET_HOLDS
MAKE_PAYMENT
MANAGE_INV
«server»
LibraryServer
+main()
ClientHandler
-clientSocket: Socket
+run()
ResponsePreparer
+prepareResponse(location: Integer, permission: PERMISSION: type: MESS_TYPE, action: ACTION, info: ArrayList<String>)
<<facade>>
RequestHandler
+processRequest(request: Message)
LibrarySystem
-name: String
+getName(): String
+getLibrary(libraryId: Integer): Library
+setName(newName): String
Library
-id: Integer
-name: String
+getId(): Integer
+getName(): String
+getInventory(): Inventory
+setName(newName: String)
AccountsDirectory
-NumAccounts
-AdminEmails: String[]
-MemberEmail: String[]
+findAccount(permissionLevel: permission, String name, Integer id)
+registerNewAccount(permission: Enum, newEmail: String, newPassword: String, newFullName: String, newBirthday: Date)
+login(inputEmail: String, inputPassword: String)
Inventory
-numMedia: Integer
+getNumMedia(): Integer
+addNewBook(newTitle: String, newAuthor: String, newPublisher: String, newGenre: String, newDdCallNumber: Double, newIsbn: String, newTotalQuanity: Integer, newTotalAvailable: Integer)
+addNewDvd(newTitle: String, newPublisher: String, newGenre: String, newRating: Rating, newDuration: Integer, newMsrp: Double, newTotalQuantity: Integer, newQuantityAvailable: Integer)
+SearchByTitle(searchTitle: String): Media
+searchByIsbn(searchIsbn: String): Book
+saveInventoryToFile(filename: String)
+loadInventoryFromFile(filename: String)
Account
-id: Integer
-email: String
-password: String
-fullName: String
-birthday: Date
-permission: Enum
+getEmail(): String
+getPassword(): String
+getName(): String
+getBirthday(): String
+setEmail(newEmail: String)
+setPassword(newPassword: String)
+setName(newName: String)
+setBirthday(Month: int, Day: int, Year: int)
+makePayment(amount: Double, ccNumber: Integer, expDate: String, cvv: Integer)
Media
-id: Integer
-title: String
-publisher: String
-genre: String
-msrp: Double
-totalQuanity: Integer
-quantityAvailable: Integer
+getId(): Integer
+getTitle(): String
+getPublisher(): String
+getGenre(): String
+getMsrp(): Double
+getTotalQuantity(): Integer
+getQuanityAvailable(): Integer
+setTitle(newTitle: String)
+setPublisher(newPublisher: String)
+setGenre(newGenre: String)
+setMsrp(newMsrp: Double)
+setTotalQuantity(newTotalQuantity: Integer)
+setQuantityAvailable(newQuantityAvailable: Integer)
Member
+Member(newEmail: String, newPassword: String, newFullName: String, newBirthday: Date):
+getLoans(): Loan[]
+getHolds(): Hold[]
Admin
+Admin(newEmail: String, newPassword: String, newFullTime: String, newBirthday: Date): Admin
+manageInventory()
+viewHistory(memberId: int)
+viewLoans(memberId: int)
+addNewAdminEmail(newEmail: String)
«server»
HoldsRepository
-holds: List<Hold>
-numHolds: Integer
+getHolds(): List<hold>
+placeHold(mediaId: Integer, memberId: Integer, until: Date): Hold
+cancelHold(mediaId: Integer, memberId: Integer): Boolean
+cancelExpiredHolds(): Integer
+saveHoldToFile(filename: String)
+loadHoldsFromFile(filename: String)
«server»
LoanRepository
-numLoans: Integer
-loanLimit: Integer
+Loan(): Loan+Loan(): Loan+Loan(): Loan+Loan(): Loan
+getNumLoans(): Integer
+getLoanLimit(): Integer
+setLoanLimit(): Integer
+getHistory(): List<Loan>
+checkoutMedia(mediaId: Integer)
+returnMedia(mediaId: Integer)
+calculateFees()
+saveLoansToFile(filename: String)
+loadLoansFromFile(filename: String)
Book
-author: String
-isbn: String
-ddCallNumber: Double
+Book(newTitle: String, newAuthor: String, newPublisher: String, newGenre: String, newDdCallNumber: Double, newIsbn: String, newTotalQuantity: Integer, newTotalAvailable: Integer)
+getAuthor(): String
+getIsbn(): String
+getDdCallNumber(): String
+setAuthor(newAuthor: String)
+setIsbn(newIsbn: String)
+setDdCallNumber(newDdCallNumber: Double)
+toString(): String
DVD
-ageRating: Rating
-runtime: int
+DVD(newTitle: String, newRating: Rating, newRuntime: int)
+getRating(): Rating
+getRuntime(): int
+setRating(newRating: Rating)
+setRuntime(newRuntime: int)
+toString(): String
BoardGame
-ageRating: Rating
-playerCountMin: int
-playerCountMax: int
-gameLength: int
+Board Game(newAgeRating: Rating, newPlayerCountMin: int, newPlayerCountMax: int, newGameLength: int): BoardGame
+getRating(): Rating
+getPlayerCountMin(): int
+getPlayerCountMax(): int
+getGameLength(): int
<<Enumeration>>
Rating
G
PG
PG-13
NC-17
R
«server»
Hold
-mediaId: Integer
-memberId: Integer
-holdUntilDate: Date
+Hold(mediaId: Integer, memberId: Integer, until: Date)
+getMediaId(): Integer
+getMemberId(): Integer
+getHoldUntilDate(): Date
+toString()
<<server>>
Loan
-mediaId: Integer
-dateCheckedOut: Date
-dateDue: Date
-dateReturned: Date
-gracePeriod: Date
+Loan(mediaId:Integer)
+getMediaId(): Integer
+setCheckoutDate()
+setReturnDate()
+setReturnDate()
+setDateDue()
+setGracePeriod()
+toString()